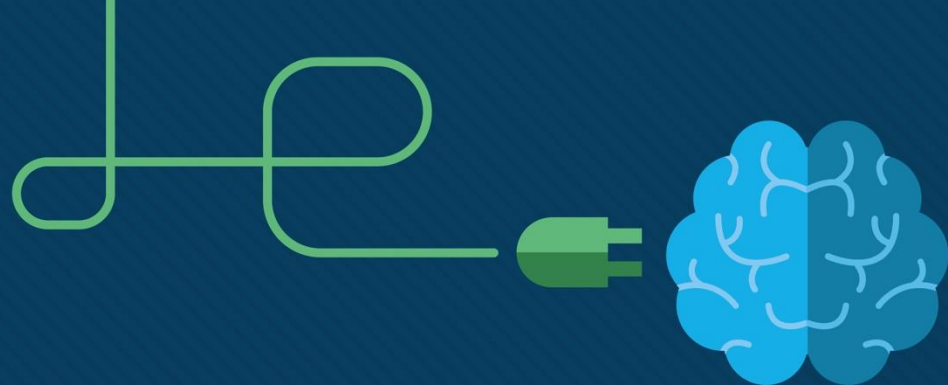




IPv6 Addressing



IPv4 Issues

IPv4 Issues

Need for IPv6

- IPv4 is running out of addresses. IPv6 is the successor to IPv4. IPv6 has a much larger 128-bit address space.
- The development of IPv6 also included fixes for IPv4 limitations and other enhancements.
- With an increasing internet population, a limited IPv4 address space, issues with NAT and the IoT, the time has come to begin the transition to IPv6.

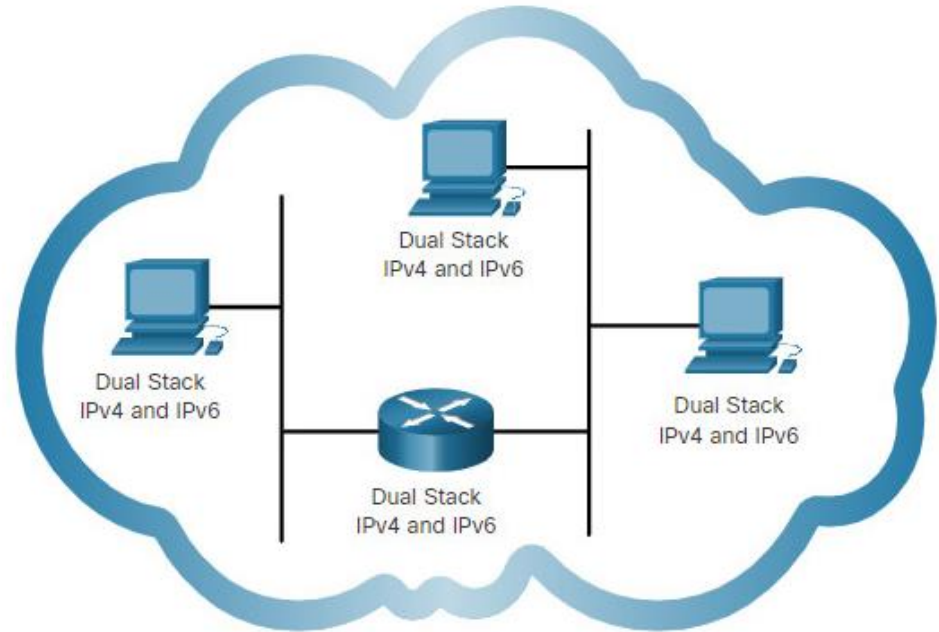


IPv4 and IPv6 Coexistence

- There is no specific date to move to IPv6. Both IPv4 and IPv6 will coexist in the near future and the transition will take several years.
- The IETF has created various protocols and tools to help network administrators migrate their networks to IPv6.
- The migration techniques can be divided into three categories:
 - Dual stack
 - Tunneling
 - Translation

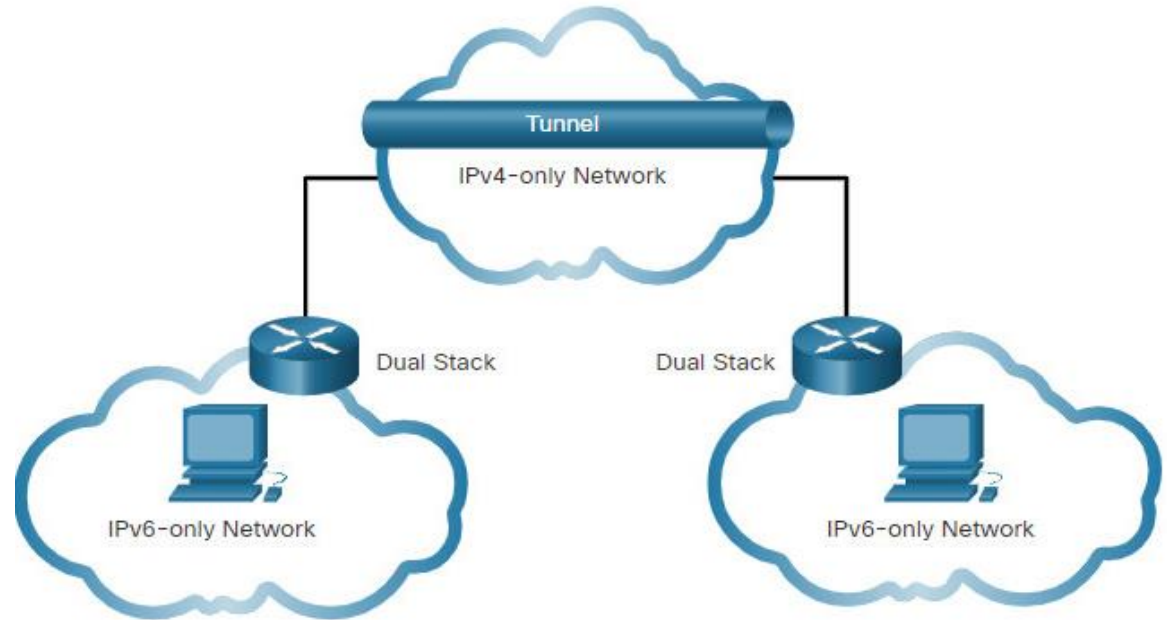
IPv4 and IPv6 Coexistence

- Dual stack allows IPv4 and IPv6 to coexist on the same network segment.
- Dual stack devices run both IPv4 and IPv6 protocol stacks simultaneously.
- Known as native IPv6, this means the customer network has an IPv6 connection to their ISP and is able to access content found on the internet over IPv6.



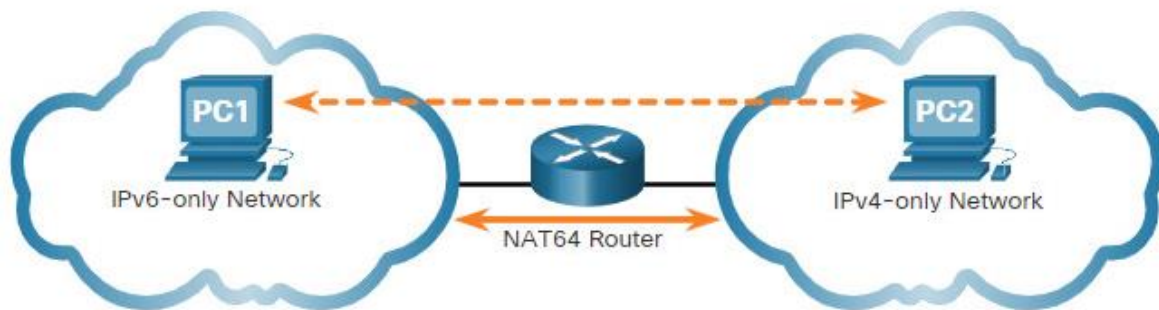
IPv4 and IPv6 Coexistence

- Tunneling is a method of transporting an IPv6 packet over an IPv4 network.
- The IPv6 packet is encapsulated inside an IPv4 packet, similar to other types of data.



IPv4 and IPv6 Coexistence

- Network Address Translation 64 (NAT64) allows IPv6-enabled devices to communicate with IPv4-enabled devices using a translation technique similar to NAT for IPv4.
- An IPv6 packet is translated to an IPv4 packet and an IPv4 packet is translated to an IPv6 packet.



IPv6 Address Representation

IPv6 Address Representation

IPv6 Addressing Formats

- IPv6 addresses are 128 bits in length and written in hexadecimal.
- IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.
- The preferred format for writing an IPv6 address is x:x:x:x:x:x:x:x, with each “x” consisting of four hexadecimal values.
- In IPv6, a hextet is the unofficial term used to refer to a segment of 16 bits, or four hexadecimal values.
- Examples of IPv6 addresses in the preferred format:

2001:0db8:0000:1111:0000:0000:0000:0200

2001:0db8:0000:00a3:abcd:0000:0000:1234

Rule 1 – Omit Leading Zero

The first rule to help reduce the notation of IPv6 addresses is to omit any leading 0s (zeros).

Examples:

- 01ab can be represented as 1ab
- 09f0 can be represented as 9f0
- 0a00 can be represented as a00
- 00ab can be represented as ab

Note: This rule only applies to leading 0s, NOT to trailing 0s, otherwise the address would be ambiguous.

Type	Format
Preferred	2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000 : 0200
No leading zeros	2001 : db8 : 0 : 1111 : 0 : 0 : 0 : 200

IPv6 Address Representation

Rule 2 – Double Colon

A double colon (::) can replace any single, contiguous string of one or more 16-bit hextets consisting of all zeros.

Example:

- 2001:db8:cafe:1:0:0:0:1 (leading 0s omitted) could be represented as 2001:db8:cafe:1::1

Note: The double colon (::) can only be used once within an address, otherwise there would be more than one possible resulting address.

Type	Format
Preferred	2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000 : 0200
Compressed	2001:db8:0:1111::200

IPv6 Address Types

IPv6 Address Types

Unicast, Multicast, Anycast

There are three broad categories of IPv6 addresses:

- **Unicast** – Unicast uniquely identifies an interface on an IPv6-enabled device.
- **Multicast** – Multicast is used to send a single IPv6 packet to multiple destinations.
- **Anycast** – This is any IPv6 unicast address that can be assigned to multiple devices. A packet sent to an anycast address is routed to the nearest device having that address.

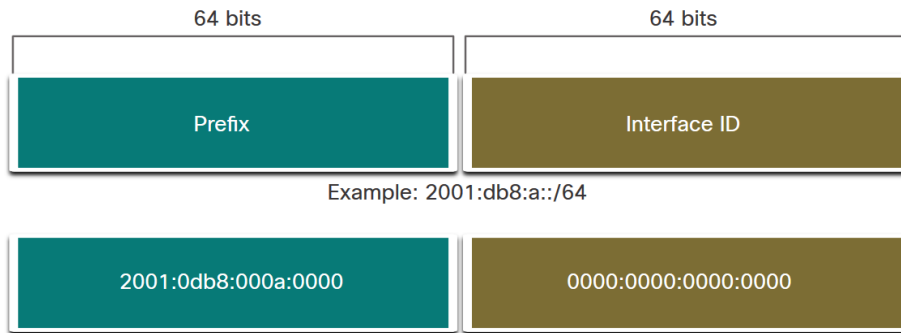
Note: Unlike IPv4, IPv6 does not have a **broadcast address**. However, there is an IPv6 **all-nodes multicast address** that essentially gives the same result.

IPv6 Address Types

IPv6 Prefix Length

Prefix length is represented in slash notation and is used to indicate the network portion of an IPv6 address.

The IPv6 prefix length can range from 0 to 128. The recommended IPv6 prefix length for LANs and most other types of networks is /64.



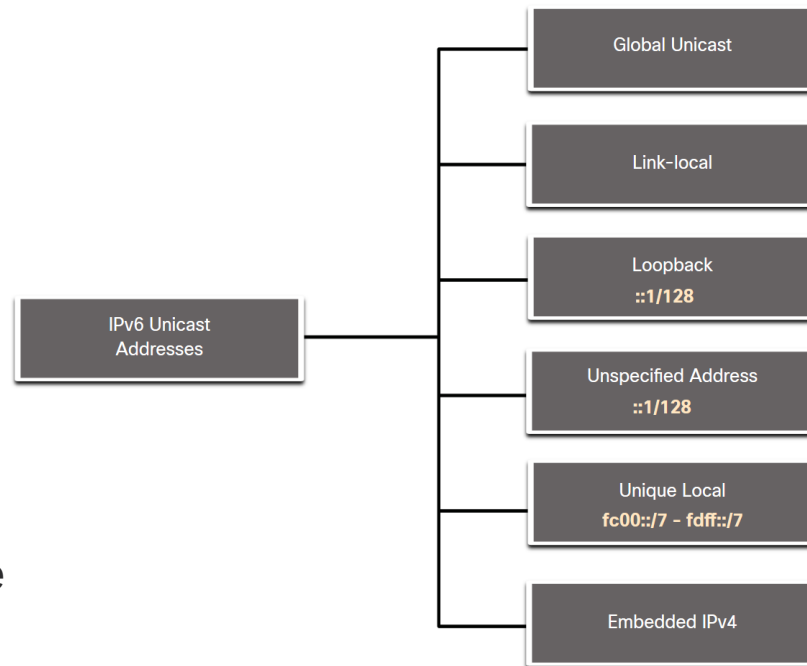
Note: It is strongly recommended to use a 64-bit Interface ID for most networks. This is because stateless address autoconfiguration (SLAAC) uses 64 bits for the Interface ID. It also makes subnetting easier to create and manage.

IPv6 Address Types

Types of IPv6 Unicast Addresses

Unlike IPv4 devices that have only a single address, IPv6 addresses typically have two unicast addresses:

- **Global Unicast Address (GUA)** – This is similar to a public IPv4 address. These are globally unique, internet-routable addresses.
- **Link-local Address (LLA)** - Required for every IPv6-enabled device and used to communicate with other devices on the same local link. LLAs are not routable and are confined to a single link.



A Note About the Unique Local Address

Unique local addresses (range fc00::/7 to fdff::/7) are not yet commonly implemented.

Unique local addresses may eventually be used to address devices that should not be accessible from the outside, such as internal servers and printers.

A Note About the Unique Local Address (Cont.)

The IPv6 unique local addresses have some similarity to RFC 1918 private addresses for IPv4, but there are significant differences:

- Unique local addresses are used for local addressing within a site or between a limited number of sites.
- Unique local addresses can be used for devices that will never need to access another network.
- Unique local addresses are not globally routed or translated to a global IPv6 address.

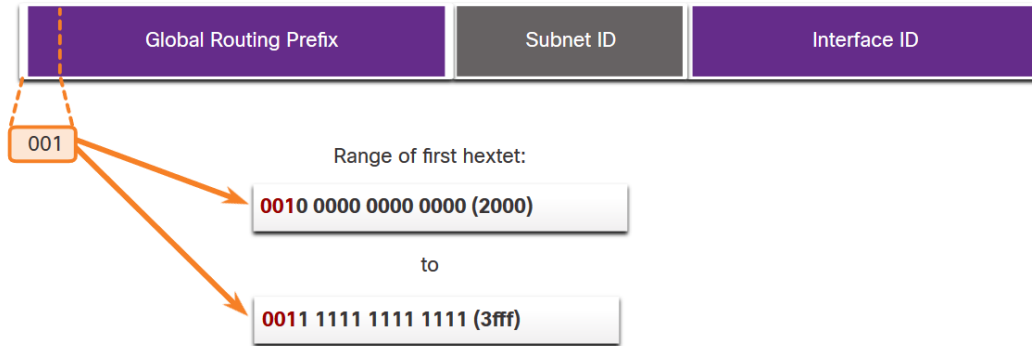
Note: Many sites use the private nature of RFC 1918 addresses to attempt to secure or hide their network from potential security risks. This was never the intended use of ULAs.

IPv6 Address Types

IPv6 GUA

IPv6 global unicast addresses (GUAs) are globally unique and routable on the IPv6 internet.

- Currently, only GUAs with the first three bits of 001 or **2000::/3** are being assigned.
- Currently available GUAs begins with a decimal 2 or a 3 (This is only 1/8th of the total available IPv6 address space).



IPv6 GUA Structure

Global Routing Prefix:

- The global routing prefix is the prefix, or network, portion of the address that is assigned by the provider, such as an ISP, to a customer or site. The global routing prefix will vary depending on ISP policies.

Subnet ID:

- The Subnet ID field is the area between the Global Routing Prefix and the Interface ID. The Subnet ID is used by an organization to identify subnets within its site.

Interface ID:

- The IPv6 interface ID is equivalent to the host portion of an IPv4 address. It is strongly recommended that in most cases /64 subnets should be used, which creates a 64-bit interface ID.

Note: IPv6 allows the all-0s and all-1s host addresses can be assigned to a device. The all-1s address can be used because broadcast addresses are not used within IPv6. The all-0s address can also be used, but is reserved as a Subnet-Router anycast address, and should be assigned only to routers.

IPv6 Address Types

IPv6 LLA

An IPv6 link-local address (LLA) enables a device to communicate with other IPv6-enabled devices on the same link and only on that link (subnet).

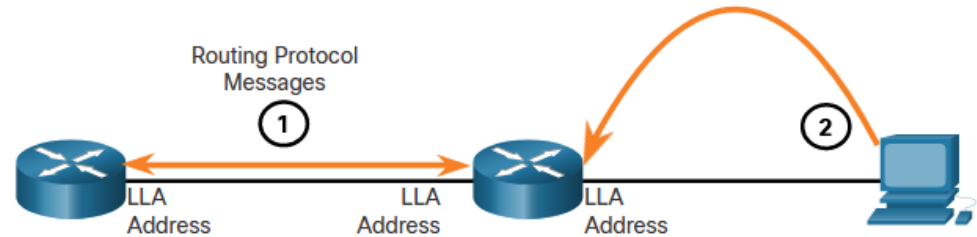
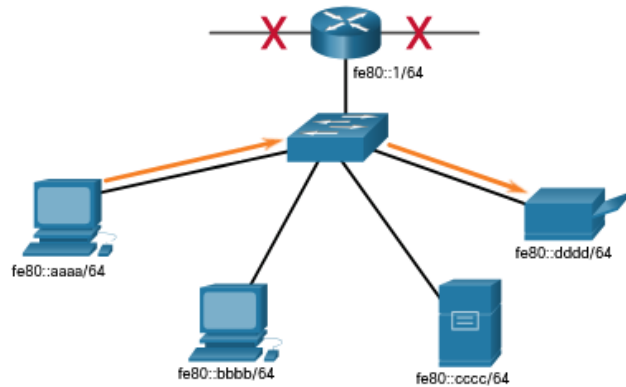
- Packets with a source or destination LLA cannot be routed.
- Every IPv6-enabled network interface must have an LLA.
- If an LLA is not configured manually on an interface, the device will automatically create its own without communicating with a DHCP server.
- IPv6 LLAs are in the **fe80::/10** range.

IPv6 Address Types

IPv6 LLA (Cont.)

IPv6 Packet

Source IPv6 Address	Destination IPv6 Address
fe80::aaaa	fe80::dddd



1. Routers use the LLA of neighbor routers to send routing updates.
2. Hosts use the LLA of a local router as the default-gateway.

GUA and LLA Static Configuration

GUA and LLA Static Configuration

Static GUA Configuration on a Router

Most IPv6 configuration and verification commands in the Cisco IOS are similar to their IPv4 counterparts. In many cases, the only difference is the use of **ipv6** in place of **ip** within the commands.

- The command to configure an IPv6 GUA on an interface is: **ipv6 address *ipv6-address/prefix-length***.
- The example shows commands to configure a GUA on the G0/0/0 interface on R1:

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
```

GUA and LLA Static Configuration

Static GUA Configuration on a Windows Host

- Manually configuring the IPv6 address on a host is similar to configuring an IPv4 address.
- The GUA or LLA of the router interface can be used as the default gateway. Best practice is to use the LLA.

Note: When DHCPv6 or SLAAC is used, the LLA of the router will automatically be specified as the default gateway address.

Internet Protocol Version 6 (TCP/IPv6) Properties

General

You can get IPv6 settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IPv6 settings.

☐ Obtain an IPv6 address automatically

☒ Use the following IPv6 address:

IPv6 address: 2001:db8:acad:1::10

Subnet prefix length: 64

Default gateway: 2001:db8:acad:1::1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

☐ Validate settings upon exit

Advanced...

OK Cancel

Static GUA Configuration of a Link-Local Unicast Address

Configuring the LLA manually lets you create an address that is recognizable and easier to remember.

- LLAs can be configured manually using the **ipv6 address** *ipv6-link-local-address* **link-local** command.
- The example shows commands to configure a LLA on the G0/0/0 interface on R1

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address fe80::1:1 link-local
R1(config-if)# no shutdown
R1(config-if)# exit
```

Note: The same LLA can be configured on each link as long as it is unique on that link. Common practice is to create a different LLA on each interface of the router to make it easy to identify the router and the specific interface.

Dynamic Addressing for IPv6 GUAs

Dynamic Addressing for IPv6 GUAs

RS and RA Messages

Devices obtain GUA addresses dynamically through Internet Control Message Protocol version 6 (ICMPv6) messages.

- Router Solicitation (RS) messages are sent by host devices to discover IPv6 routers
- Router Advertisement (RA) messages are sent by routers to inform hosts on how to obtain an IPv6 GUA and provide useful network information such as:
 - Network prefix and prefix length - This tells the device which network it belongs to.
 - Default gateway address - This is an IPv6 LLA, the source IPv6 address of the RA message.
 - DNS addresses and domain name - These are the addresses of DNS servers and a domain name.

Dynamic Addressing for IPv6 GUAs

RS and RA Messages (Cont.)

- The RA can provide three methods for configuring an IPv6 GUA :
 - SLAAC - “I have everything you need including the prefix, prefix length, and default gateway address.”
 - SLAAC with stateless DHCPv6 server - “Here is my information but you need to get other information such as DNS addresses from a stateless DHCPv6 server.”
 - Stateful DHCPv6 (no SLAAC) - “I can give you your default gateway address. You need to ask a stateful DHCPv6 server for all your other information.”

Dynamic Addressing for IPv6 GUAs

Method 1: SLAAC

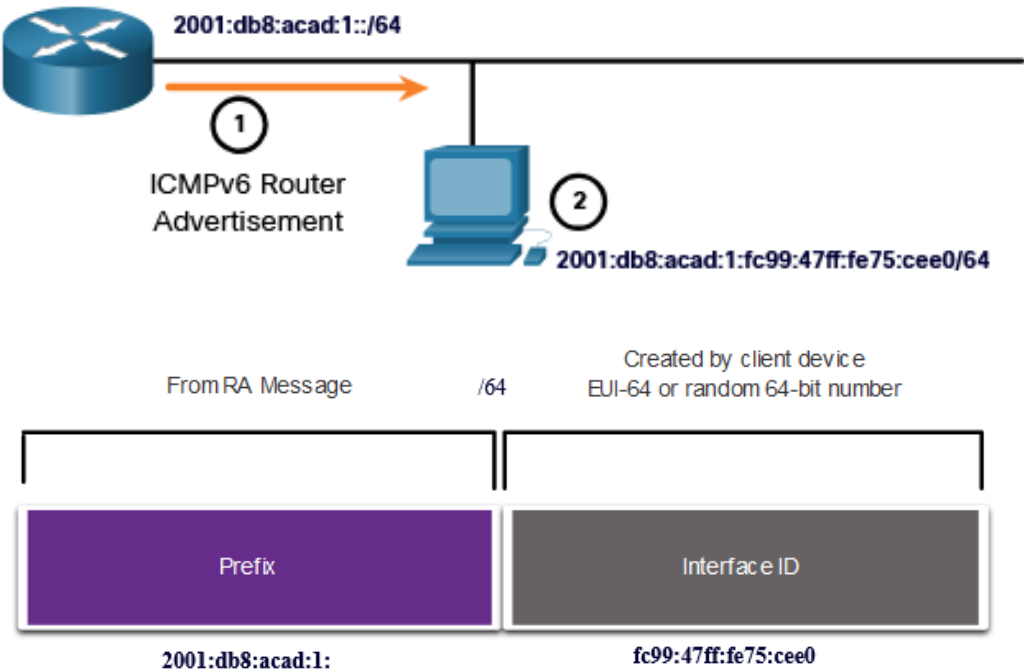
- SLAAC allows a device to configure a GUA without the services of DHCPv6.
- Devices obtain the necessary information to configure a GUA from the ICMPv6 RA messages of the local router.
- Two parts of the address are created as follows:

Prefix - This is advertised in the RA message.

Interface ID - This uses the EUI-64 process or by generating a random 64-bit number, depending on the device operating system.

Dynamic Addressing for IPv6 GUAs

Method 1: SLAAC



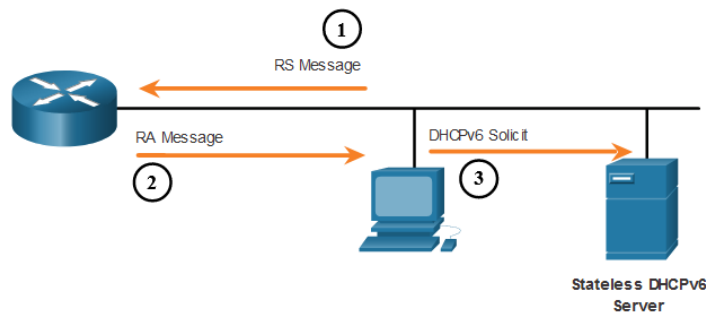
- 1.The router sends an RA message with the prefix for the local link.
- 2.The PC uses SLAAC to obtain a prefix from the RA message and creates its own Interface ID.

Method 2: SLAAC and Stateless DHCP

An RA can instruct a device to use both SLAAC and stateless DHCPv6.

The RA message suggests devices use the following:

- SLAAC to create its own IPv6 GUA
- The router LLA, which is the RA source IPv6 address, as the default gateway address
- A stateless DHCPv6 server to obtain other information such as a DNS server address and a domain name



Dynamic Addressing for IPv6 GUAs

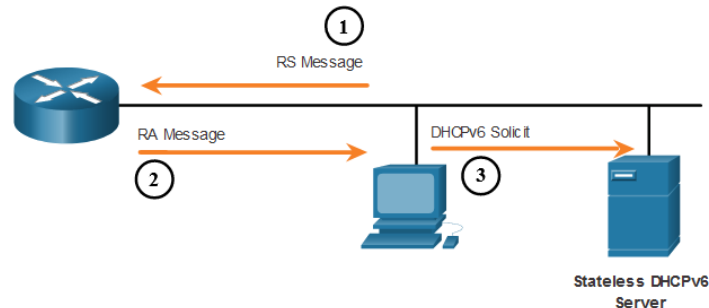
Method 3: Stateful DHCPv6

An RA can instruct a device to use stateful DHCPv6 only.

Stateful DHCPv6 is similar to DHCP for IPv4. A device can automatically receive a GUA, prefix length, and the addresses of DNS servers from a stateful DHCPv6 server.

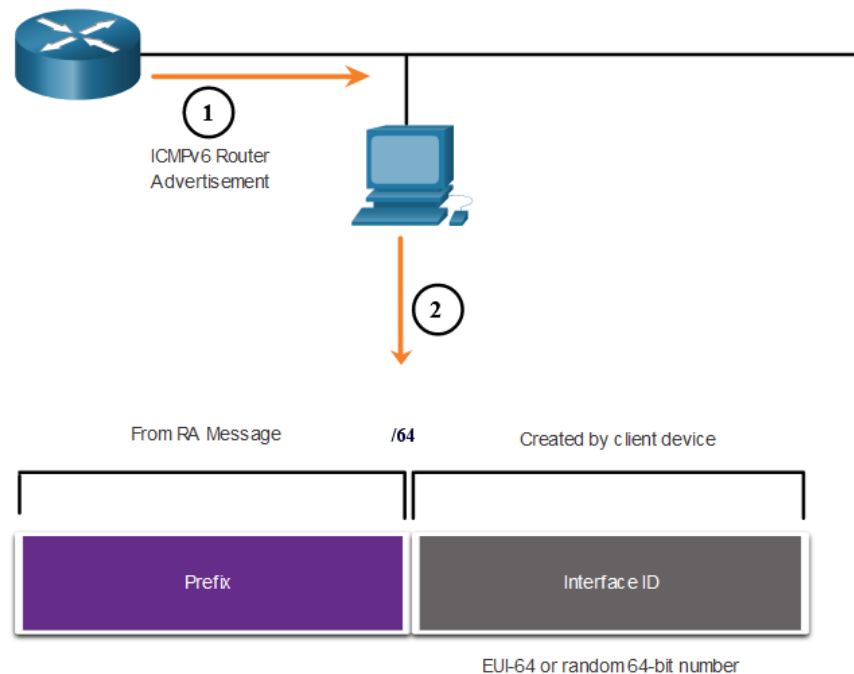
The RA message suggests devices use the following:

- The router LLA, which is the RA source IPv6 address, for the default gateway address.
- A stateful DHCPv6 server to obtain a GUA, DNS server address, domain name and other necessary information.



EUI-64 Process vs. Randomly Generated

- When the RA message is either SLAAC or SLAAC with stateless DHCPv6, the client must generate its own interface ID.
- The interface ID can be created using the EUI-64 process or a randomly generated 64-bit number.



Dynamic Addressing for IPv6 GUAs

EUI-64 Process

The IEEE defined the **Extended Unique Identifier (EUI)** or modified EUI-64 process.

This process uses the 48-bit Ethernet MAC address of a client, and inserts another 16 bits in the middle of the 48-bit MAC address to create a 64-bit interface ID.

An EUI-64 Interface ID is represented in binary and is made up of three parts:

- 24-bit OUI from the client MAC address, but the 7th bit (the Universally/Locally (U/L) bit) is reversed. This means that if the 7th bit is a 0, it becomes a 1, and vice versa.
- The inserted 16-bit value fffe (in hexadecimal).
- 24-bit Device Identifier from the client MAC address.
- Example:

48-bit MAC	fc:99:47:75:ce:e0
EUI-64 Interface ID	fe:99:47:ff:fe:75:ce:e0

Dynamic Addressing for IPv6 GUAs

Randomly Generated Interface IDs

Depending upon the operating system, a device may use a randomly generated interface ID instead of using the MAC address and the EUI-64 process.

Beginning with Windows Vista, Windows uses a randomly generated interface ID instead of one created with EUI-64.

Windows XP and previous Windows operating systems used EUI-64.

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2001:db8:acad:1:50a5:8a35:a5bb:66e1
Link-local IPv6 Address . . . . . : fe80::50a5:8a35:a5bb:66e1
Default Gateway . . . . . : fe80::1
C:\>
```

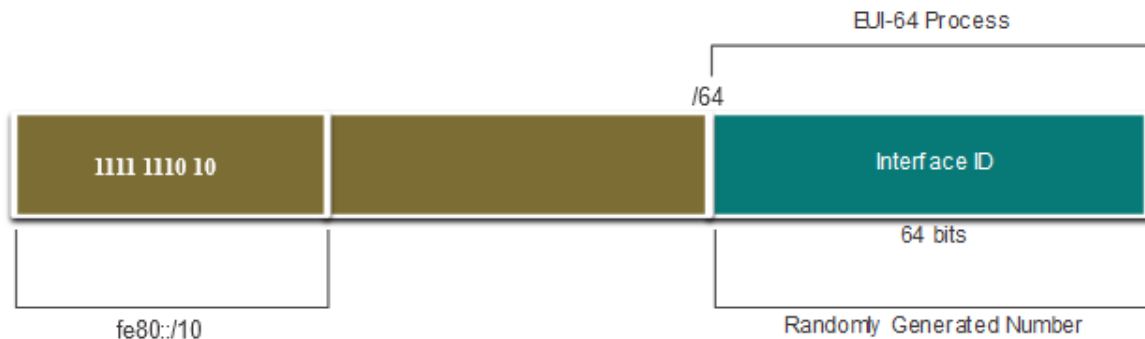
Note: To ensure the uniqueness of any IPv6 unicast address, the client may use a process known as **Duplicate Address Detection (DAD)**. This is similar to an ARP request for its own address. If there is no reply, then the address is unique.

Dynamic Addressing for IPv6 LLAs

Dynamic Addressing for IPv6 LLAs

Dynamic LLAs

- All IPv6 interfaces must have an IPv6 LLA.
- Like IPv6 GUAs, LLAs can be configured dynamically.
- The figure shows the LLA is dynamically created using the **fe80::/10 prefix** and the interface ID using the EUI-64 process, or a randomly generated 64-bit number.



Dynamic Addressing for IPv6 LLAs

Dynamic LLAs on Windows

Operating systems, such as Windows, will typically use the same method for both a SLAAC-created GUA and a dynamically assigned LLA.

EUI-64 Generated Interface ID:

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2001:db8:acad:1:fc99:47ff:fe75:cee0
Link-local IPv6 Address . . . . . : fe80::fc99:47ff:fe75:cee0
Default Gateway . . . . . : fe80::1
C:\>
```

Random 64-bit Generated Interface ID:

```
C:\> ipconfig
Windows IP Configuration
Ethernet adapter Local Area Connection:
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2001:db8:acad:1:50a5:8a35:a5bb:66e1
Link-local IPv6 Address . . . . . : fe80::50a5:8a35:a5bb:66e1
Default Gateway . . . . . : fe80::1
C:\>
```

Dynamic Addressing for IPv6 LLAs

Dynamic LLAs on Cisco Routers

Cisco routers automatically create an IPv6 LLA whenever a GUA is assigned to the interface. By default, Cisco IOS routers use EUI-64 to generate the interface ID for all LLAs on IPv6 interfaces.

Here is an example of a LLA dynamically configured on the G0/0/0 interface of R1:

```
R1# show interface gigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Hardware is ISR4221-2x1GE, address is 7079.b392.3640 (bia 7079.b392.3640)
(Output omitted)
R1# show ipv6 interface brief
GigabitEthernet0/0/0 [up/up]
FE80::7279:B3FF:FE92:3640
2001:DB8:ACAD:1::1
```

Dynamic Addressing for IPv6 LLAs

Verify IPv6 Address Configuration

```
R1# show ipv6 interface brief
GigabitEthernet0/0/0    [up/up]
    FE80::7279:b3ff:fe92:3640
    2001:DB8:ACAD:1::1
GigabitEthernet0/0/1    [up/up]
    FE80::8320:e4ff:fe79:2830
    2001:DB8:ACAD:2::1
Serial0/1/0              [up/up]
    FE80::7279:b3ff:fe92:3640
    2001:DB8:ACAD:3::1
Serial0/1/1              [down/down]
    unassigned
R1#
```

- The `show ipv6 interface brief` command displays the IPv6 address of the Ethernet interfaces.
- EUI-64 uses this MAC address to generate the interface ID for the LLA.
- Additionally, the `show ipv6 interface brief` command displays abbreviated output for each of the interfaces.
- The `[up/up]` output on the same line as the interface indicates the Layer 1/Layer 2 interface state.
- This is the same as the Status and Protocol columns in the equivalent IPv4 command.

Verify IPv6 Address Configuration (Cont.)

```
R1# show ipv6 route
IPv6 Routing Table - default - 7 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

C   2001:DB8:ACAD:1::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:1::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
C   2001:DB8:ACAD:2::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:2::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:3::/64 [0/0]
    via Serial0/1/0, directly connected
L   2001:DB8:ACAD:3::1/128 [0/0]
    via Serial0/1/0, receive
L   FF00::/8 [0/0]
    via Null0, receive
```

- show ipv6 route command can be used to verify that IPv6 networks and specific IPv6 interface addresses have been installed in the IPv6 routing table.
- The show ipv6 route command will only display IPv6 networks, not IPv4 networks.

Verify IPv6 Address Configuration (Cont.)

```
R1# ping 2001:db8:acad:1::10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:1::10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#
```

- The ping command for IPv6 is identical to the command used with IPv4, except that an IPv6 address is used.
- As shown in the example, the command is used to verify Layer 3 connectivity between R1 and PC1.
- When pinging an LLA from a router, Cisco IOS will prompt the user for the exit interface.
- Because the destination LLA can be on one or more of its links or networks, the router needs to know which interface to send the ping to.

IPv6 Multicast Addresses

IPv6 Multicast Addresses

Assigned IPv6 Multicast Addresses

Pv6 multicast addresses are similar to IPv4 multicast addresses. IPv6 multicast addresses have the prefix `ff00::/8`. There are two types of IPv6 multicast addresses:

- Well-Known multicast addresses
- Solicited node multicast addresses

Note: Multicast addresses can only be destination addresses and not source addresses.

Well-Known IPv6 Multicast Addresses

Assigned multicast addresses are reserved multicast addresses for predefined groups of devices. An assigned multicast address is a single address used to reach a group of devices running a common protocol or service. Assigned multicast addresses are used in context with specific protocols such as DHCPv6.

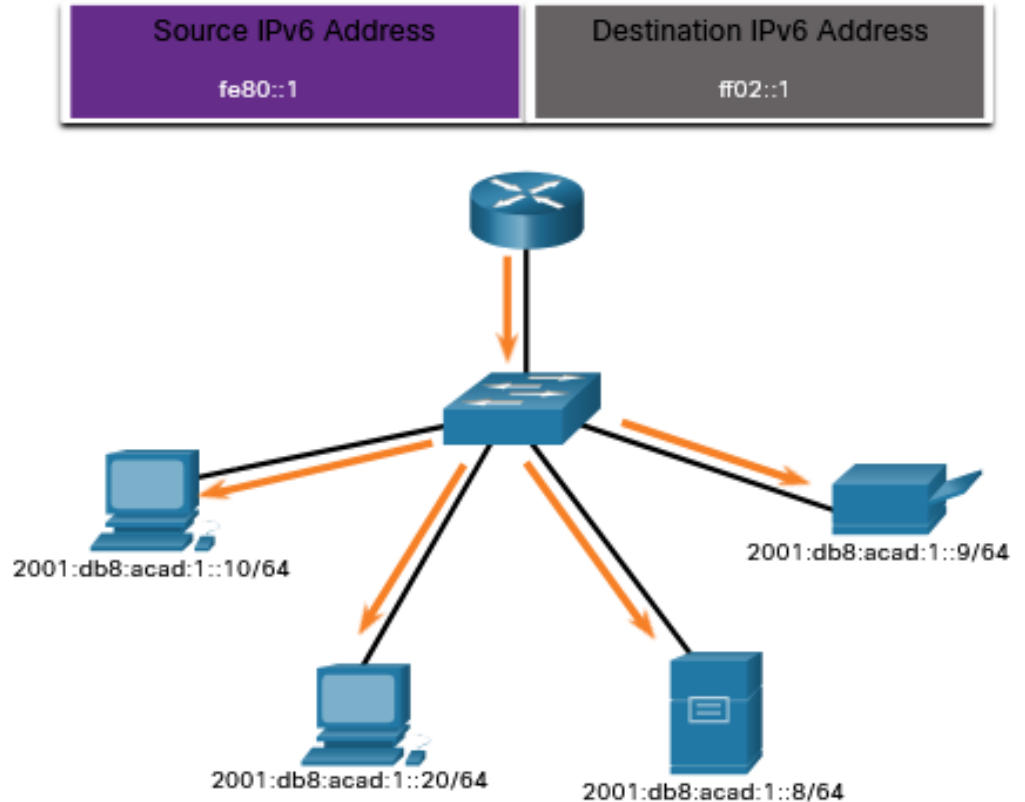
There are two common IPv6 Assigned multicast groups:

- **ff02::1 All-nodes multicast group** - This is a multicast group that all IPv6-enabled devices join. A packet sent to this group is received and processed by all IPv6 interfaces on the link or network.
- **ff02::2 All-routers multicast group** - This is a multicast group that all IPv6 routers join. A router becomes a member of this group when it is enabled as an **IPv6 router with the ipv6 unicast-routing** global configuration command. A packet sent to this group is received and processed by all IPv6 routers on the link or network.

IPv6 Multicast Addresses

IPv6 All-Nodes Multicast: RA Message

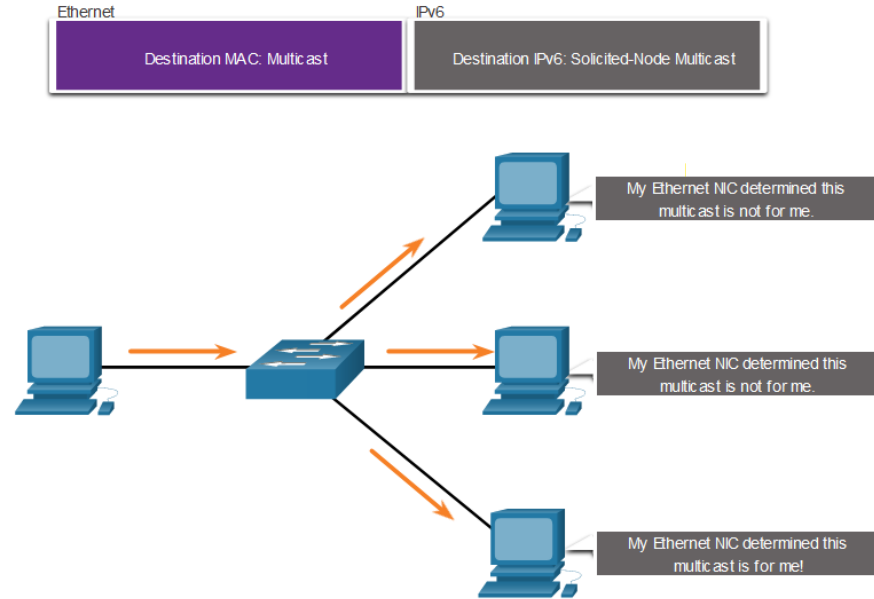
- IPv6-enabled devices send ICMPv6 RS messages to the all-routers multicast address.
- The RS message requests an RA message from the IPv6 router to assist the device in its address configuration.
- The IPv6 router responds with an RA message, as shown.



IPv6 Multicast Addresses

Solicited-Node IPv6 Multicast

- A solicited-node multicast address is similar to the all-nodes multicast address.
- A solicited-node multicast address is mapped to a special Ethernet multicast address.
- The Ethernet NIC can filter the frame by examining the destination MAC address without sending it to the IPv6 process to see if the device is the intended target of the IPv6 packet.



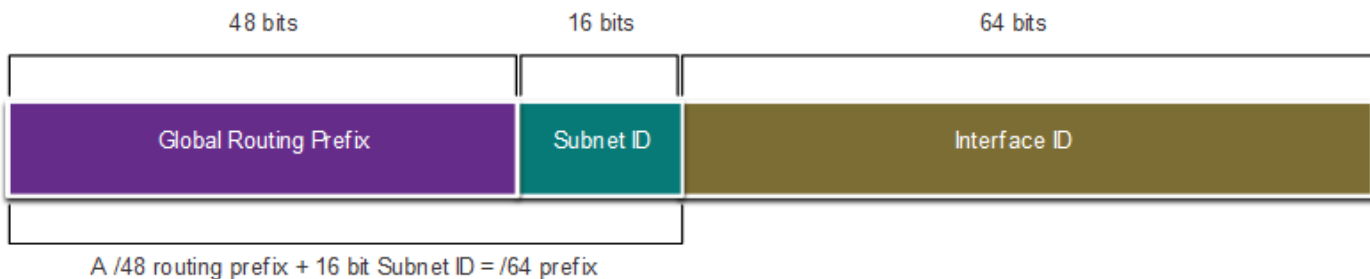
Subnet an IPv6 Network

Subnet an IPv6 Network

Subnet Using the Subnet ID

IPv6 was designed with subnetting in mind.

- A separate subnet ID field in the IPv6 GUA is used to create subnets.
- The subnet ID field is the area between the Global Routing Prefix and the interface ID.



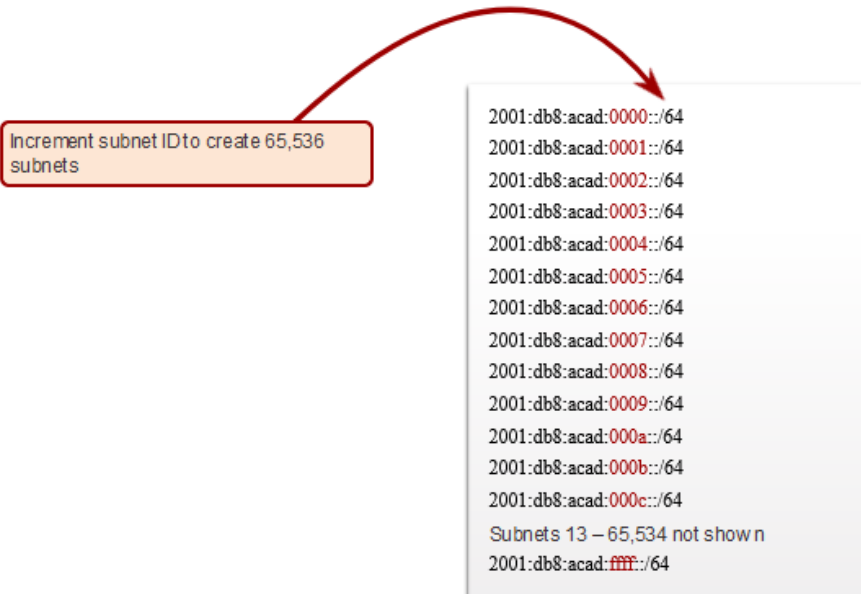
Subnet an IPv6 Network

IPv6 Subnetting Example

Given the 2001:db8:acad::/48 global routing prefix with a 16 bit subnet ID.

- Allows 65,536 /64 subnets
- The global routing prefix is the same for all subnets.
- Only the subnet ID hextet is incremented in hexadecimal for each subnet.

Increment subnet ID to create 65,536 subnets



```
2001:db8:acad:0000::/64
2001:db8:acad:0001::/64
2001:db8:acad:0002::/64
2001:db8:acad:0003::/64
2001:db8:acad:0004::/64
2001:db8:acad:0005::/64
2001:db8:acad:0006::/64
2001:db8:acad:0007::/64
2001:db8:acad:0008::/64
2001:db8:acad:0009::/64
2001:db8:acad:000a::/64
2001:db8:acad:000b::/64
2001:db8:acad:000c::/64
Subnets 13 – 65,534 not shown
2001:db8:acad:fff::/64
```

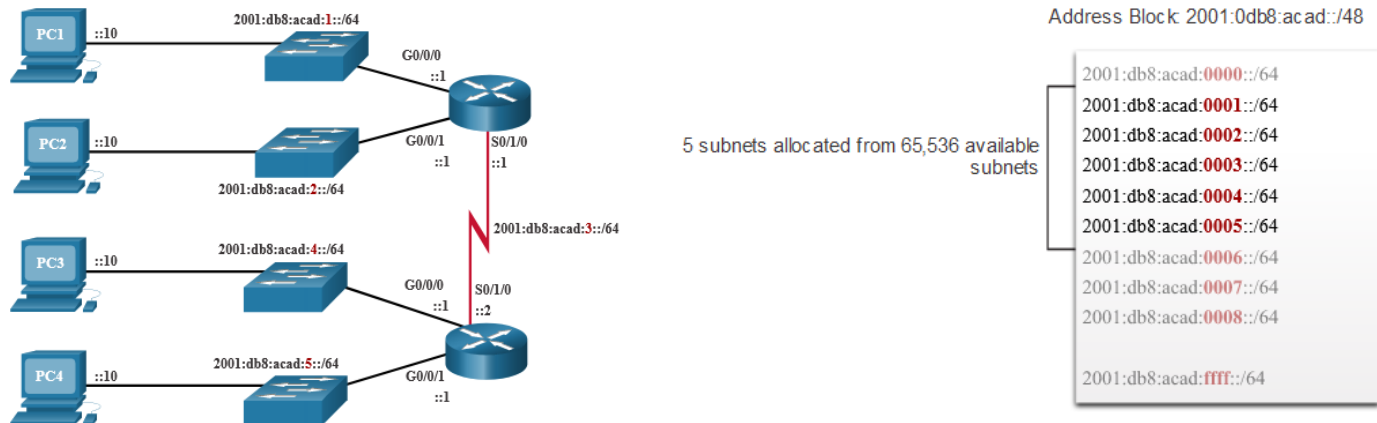
Subnet an IPv6 Network

IPv6 Subnet Allocation

The example topology requires five subnets, one for each LAN as well as for the serial link between R1 and R2.

Unlike the example for IPv4, with IPv6 the serial link subnet will have the same prefix length as the LANs. Although this may seem to “waste” addresses, address conservation is not a concern when using IPv6.

In the following figure, the five IPv6 subnets were allocated, with the subnet ID field 0001 through 0005. Each /64 subnet will provide more addresses than will ever be needed.



Subnet an IPv6 Network

Router Configured with IPv6 Subnets

The example shows that each of the router interfaces on R1 has been configured to be on a different IPv6 subnet.

```
R1(config)# interface gigabitethernet 0/0/0
R1(config-if)# ipv6 address 2001:db8:acad:1::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface gigabitethernet 0/0/1
R1(config-if)# ipv6 address 2001:db8:acad:2::1/64
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface serial 0/1/0
R1(config-if)# ipv6 address 2001:db8:acad:3::1/64
R1(config-if)# no shutdown
```

